Week 06 – Lab Assignment

Objective: To demonstrate core file operations, including creating, reading, appending, and deleting text files.

1. Python File Access Modes

Understanding how to open a file is the first step. Below are the primary modes used in Python:

Mode	Name	Description
'w'	Write	Creates a new file or overwrites an existing one.
'r'	Read	Opens an existing file for reading (Default).
'a'	Append	Adds data to the end of the file without overwriting.
'x'	Exclusive	Creates a new file; fails if the file already exists.

2. Implementation: TXT File Operations

The following script demonstrates the full lifecycle of a text file using both the standard open() method and the more secure with statement (Context Manager).

```python
import os

# --- Step 1: Create and Write to a File ---
print("=== Step 1: Writing to File ===")
# 'w' mode: creates a new file or overwrites existing content
file = open("student_data.txt", "w")
file.write("Name: Ali\n")
file.write("Class: 10\n")
file.write("Marks: 85\n")
file.close()
print("File created and data written successfully!\n")

# --- Step 2: Read Data from the File ---
print("=== Step 2: Reading from File ===")
file = open("student_data.txt", "r")
content = file.read()
print("Current Content:")
print("-" * 15)
print(content)
file.close()

# --- Step 3: Append More Data to the File ---
print("=== Step 3: Appending to File ===")
# 'a' mode: adds data to the end of the file
file = open("student_data.txt", "a")
file.write("Grade: A\n")
file.close()
print("New data 'Grade: A' appended successfully!\n")

# --- Step 4: Secure Reading (Using 'with' statement) ---
print("=== Step 4: Reading After Append (using Context Manager) ===")
# The 'with' statement automatically closes the file for us
with open("student_data.txt", "r") as file:
    print(file.read())

# --- Step 5: Delete File ---
print("=== Step 5: File Deletion ===")
if os.path.exists("student_data.txt"):
    os.remove("student_data.txt")
    print("File 'student_data.txt' deleted successfully.")
else:
    print("Error: File not found.")
```

Output:

```
C:\Python314\python.exe "D:\dev\Py - PR\Labs\Lab Assignments\main.py"
=== Step 1: Writing to File ===
File created and data written successfully!

=== Step 2: Reading from File ===
Current Content:
---------------
Name: Ali
Class: 10
Marks: 85

=== Step 3: Appending to File ===
New data 'Grade: A' appended successfully!

=== Step 4: Reading After Append (using Context Manager) ===
Name: Ali
Class: 10
Marks: 85
Grade: A

=== Step 5: File Deletion ===
File 'student_data.txt' deleted successfully.

Process finished with exit code 0
```